

Vitamins , Minerals and water

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Vitamin Facts

- Vitamins are essential **organic nutrients**, required in **small amounts**.
- Mostly obtained from out diet, bacteria & sun.
- Required for growth, maintenance, reproduction and lactation.

History

- ❖ Polish biochemist Casimir Funk discovered vitamin B1 in 1912 in rice bran.
- ❖ He proposed the complex be named "Vitamin" (vital amines).
- ❖ By the time it was shown that not all vitamins were *amines*, the word was already ubiquitous.

Classes of Vitamins

Fat Soluble Vitamins:

stored in tissues

□ A

□ D

□ E

□ K

Water Soluble Vitamins:

not stored in tissues, must have constant supply

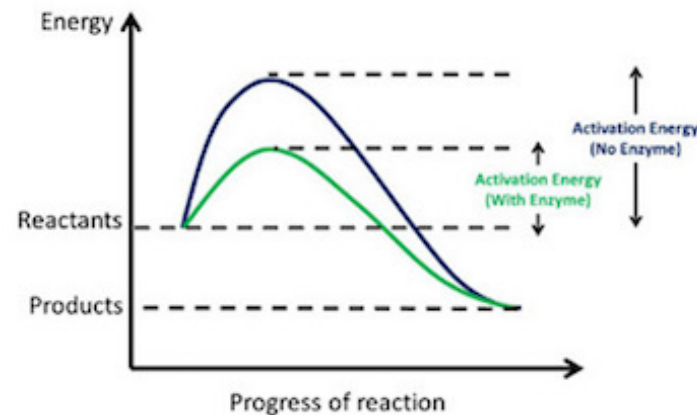
- Vitamin B₁ (thiamine)
- Vitamin B₂ (riboflavin)
- Vitamin B₃ or Vitamin P or Vitamin PP (niacin)
- Vitamin B₅ (panthotenic acid)
- Vitamin B₆ (pyridoxine and pyridoxamine)
- Vitamin B₇ or Vitamin H (biotin)
- Vitamin B₉ or Vitamin M and Vitamin B-c (folic acid)
- Vitamin B₁₂ (cobalamin)
- Vitamin C (Ascorbic acid)

Vitamins & Minerals: What For ?

Enzymes: Biological catalyst that speed up reaction and processes in the body.
They remain unchanged after the reaction.

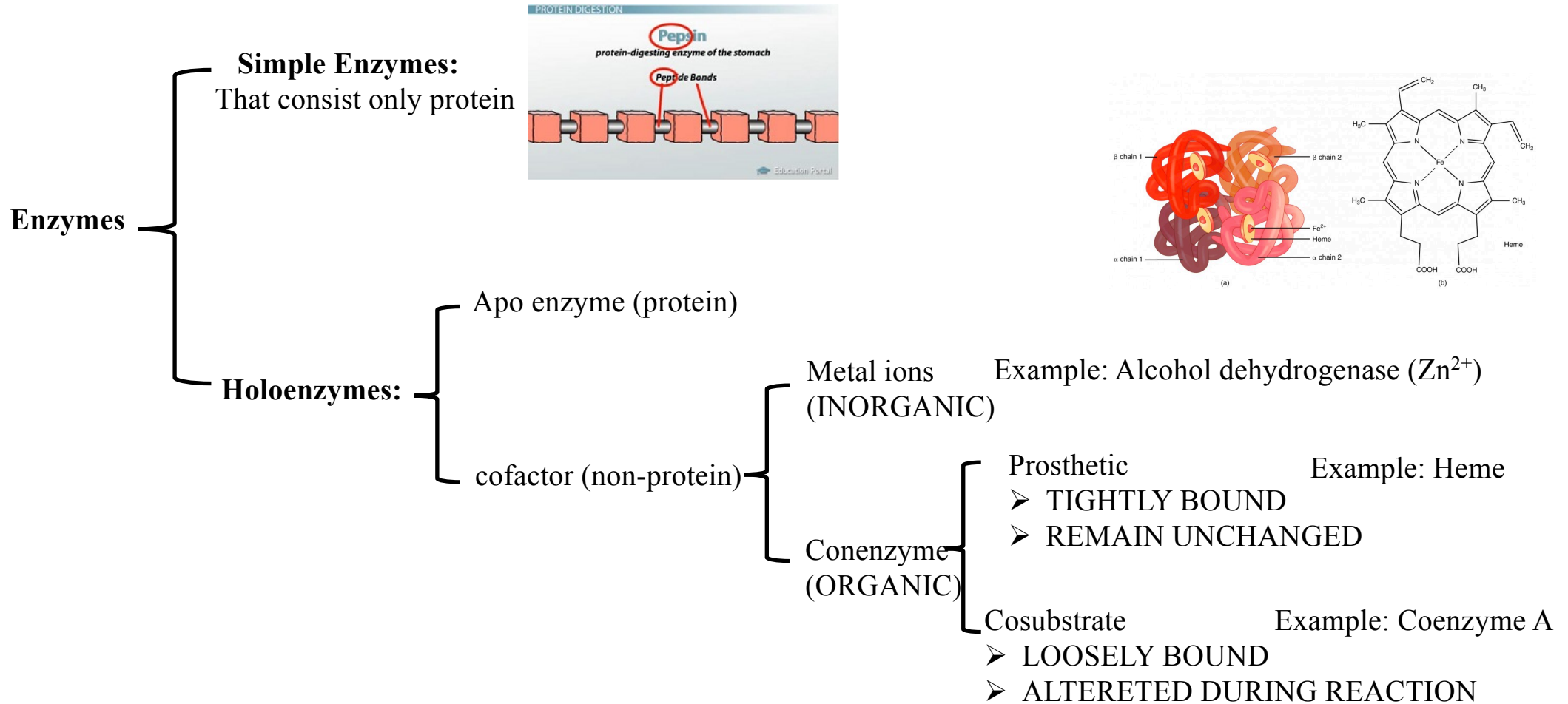
How ?

Enzymes reduce the activation energy of a reaction or process.



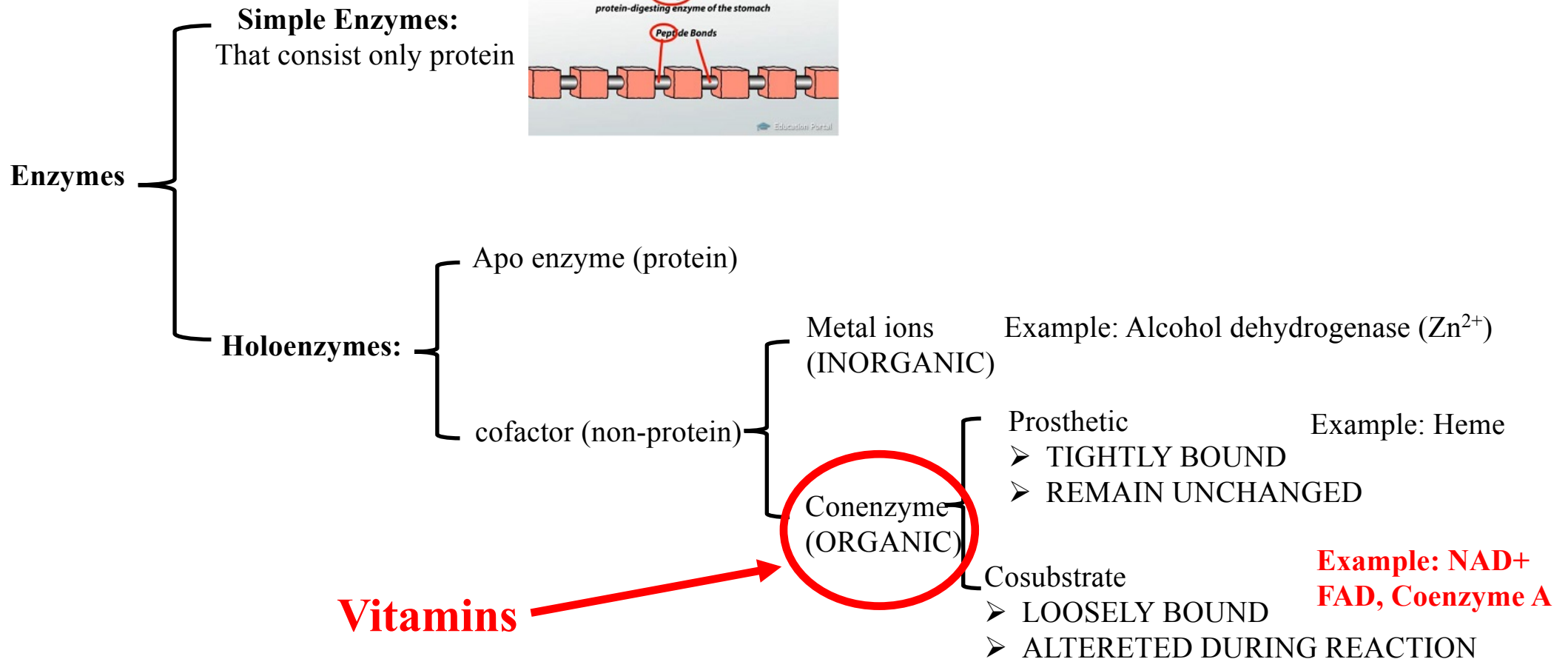
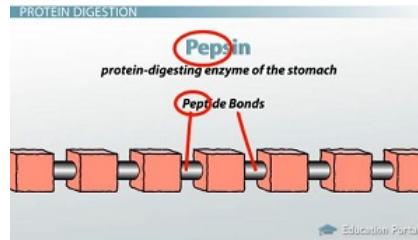
Enzymes:

Many enzymes are proteins (NOT ALL, RNA can also acts as enzyme)



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Three Vitamins produces key players
for energy production (ATP)

Vitamin B₂ → FAD

Vitamin B₃ → NAD⁺

Vitamin B₅ → CoA

**Redox
reaction**

NAD⁺, FAD, CoA ??

**Reduced
coenzyme**

**Don't
FORGET!**



NAD⁺

$2e + H^+$

NADH

FAD

$2e + 2H^+$

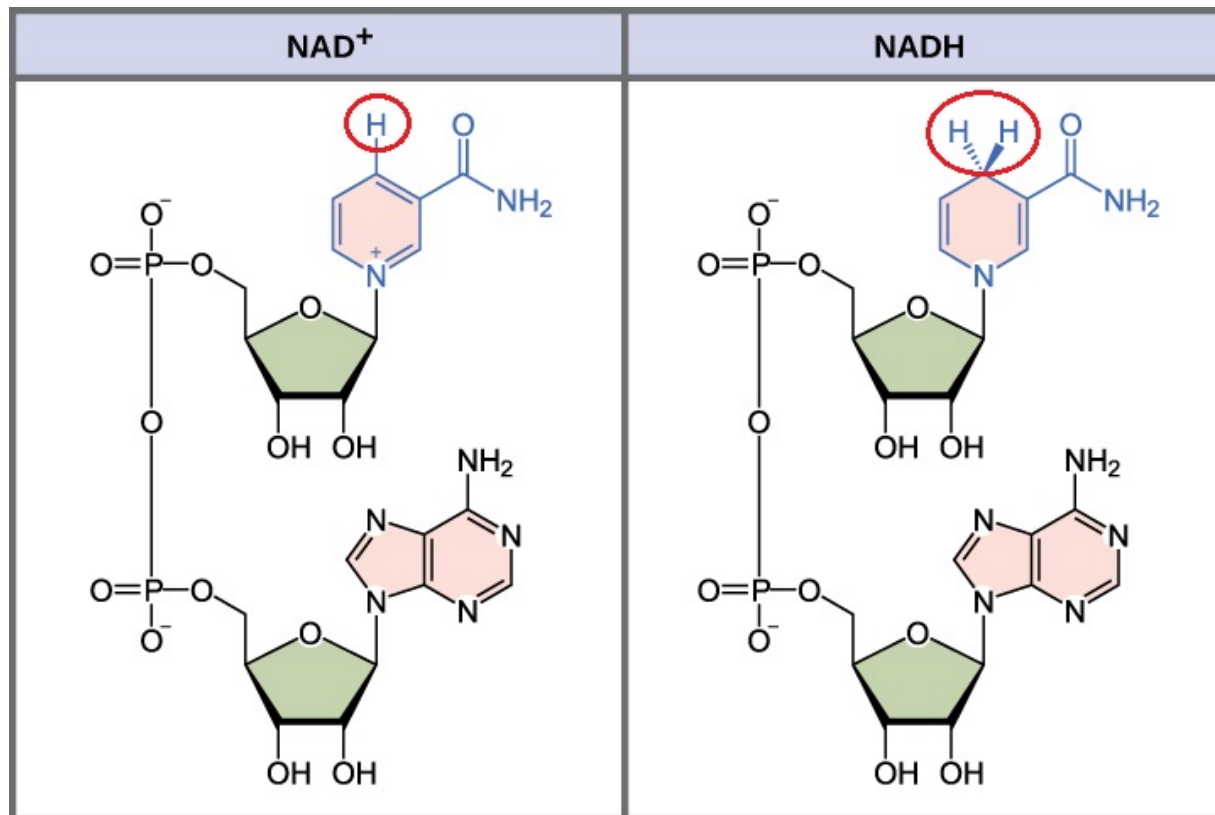
FADH₂

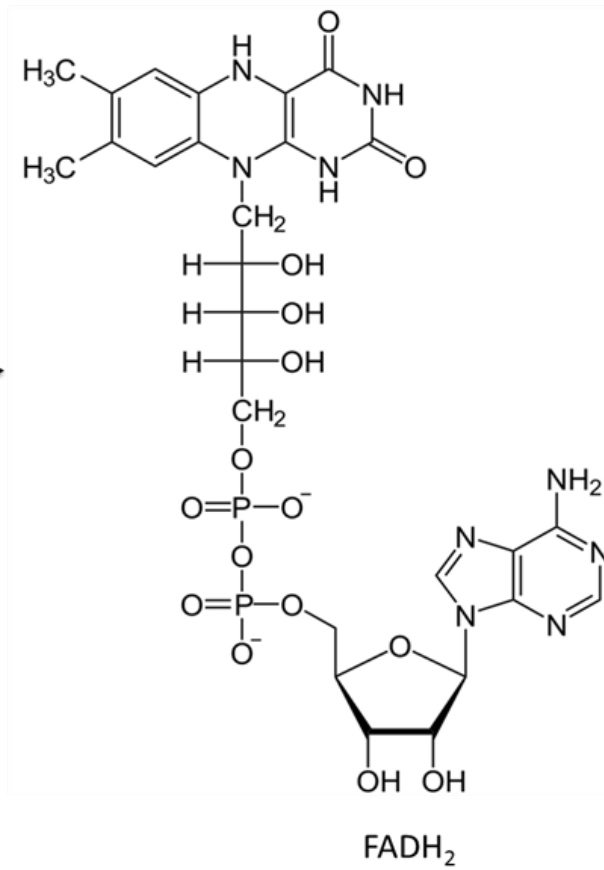
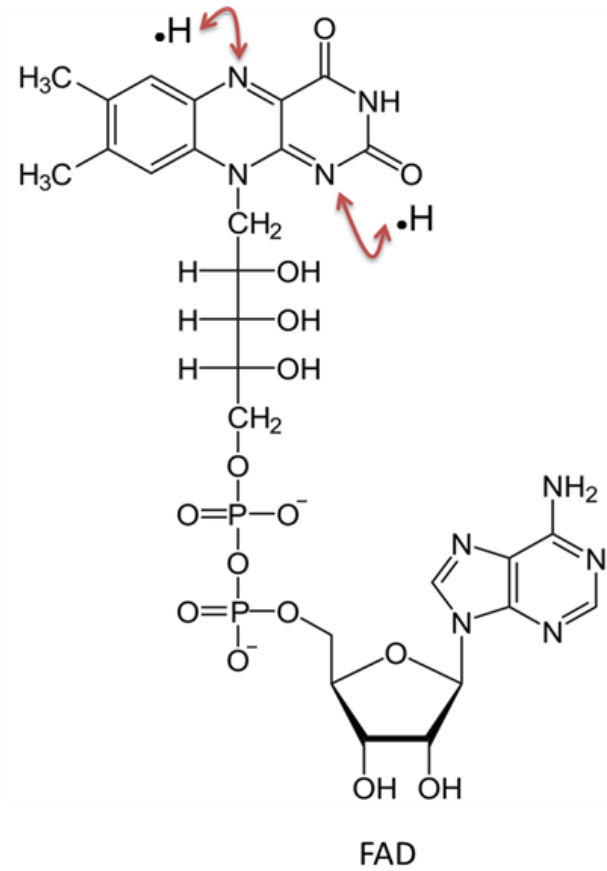
CoA

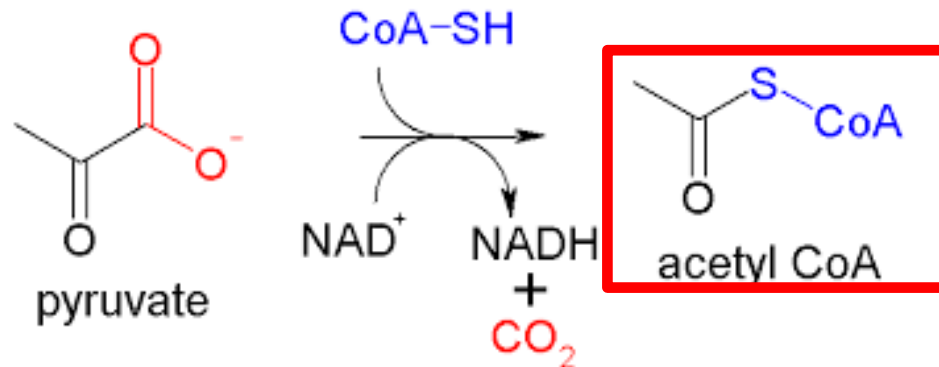
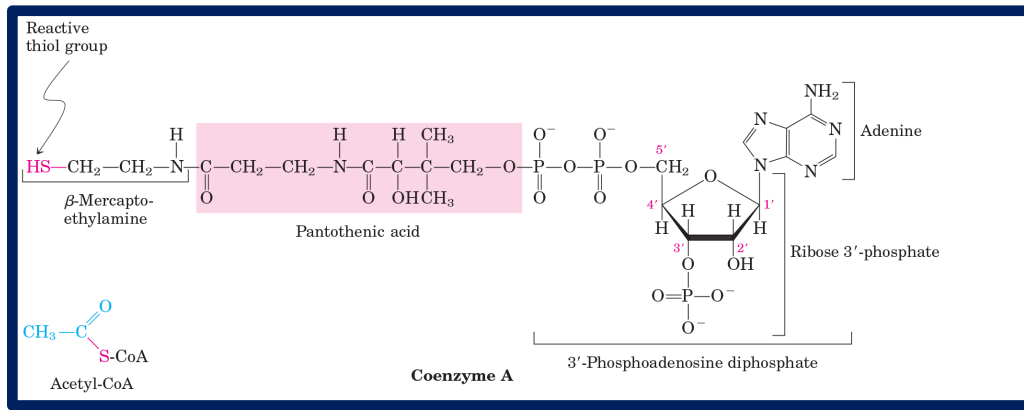
Acetyl group

Acetyl-CoA

Transfer acetyl Group







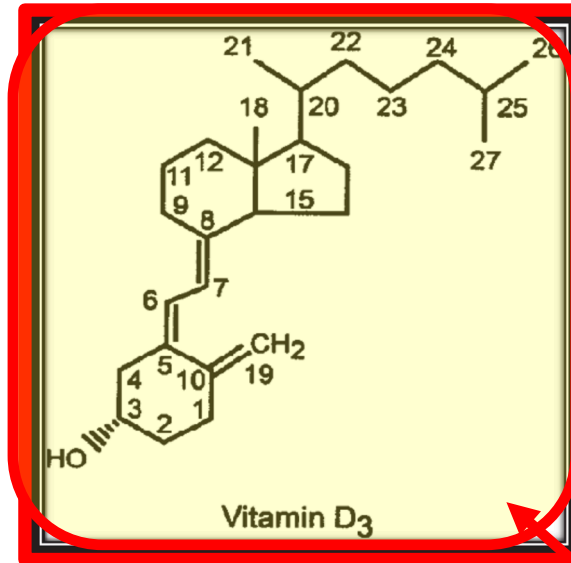
Fat/lipid Soluble Vitamins:

☐ A

☒ D

☐ E

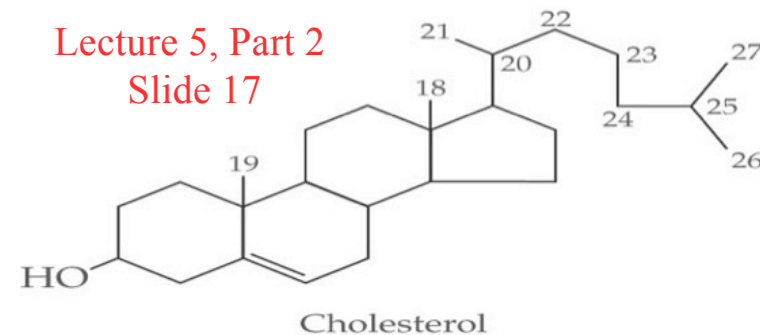
☐ K



Vitamin D can be synthesized in adequate amounts by most mammals if exposed to sufficient sunlight, it is not essential, so technically not a vitamin

1. Fat soluble (Can be stored)
2. Important for Calcium metabolism
3. Gene expression and Immune system
4. Stored in the Liver
5. Synthesized from CHOLESTEROL

Lecture 5, Part 2
Slide 17



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Minerals (Chemical element)

Macrominerals

- Calcium (Ca)
- Phosphorous (P)
- Potassium (K)
- Sulfur (S)
- Sodium (Na)
- Chlorine (Cl)
- Magnesium (Mg)

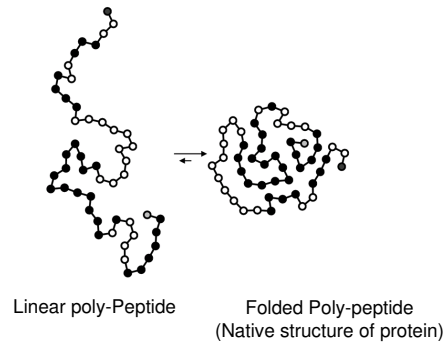
Microminerals

- Iodine (I)
- Copper (Cu)
- Iron (Fe)
- Selenium (Se)
- Manganese (Mn)
- Molybdenum (Mo)
- Zinc (Zn)

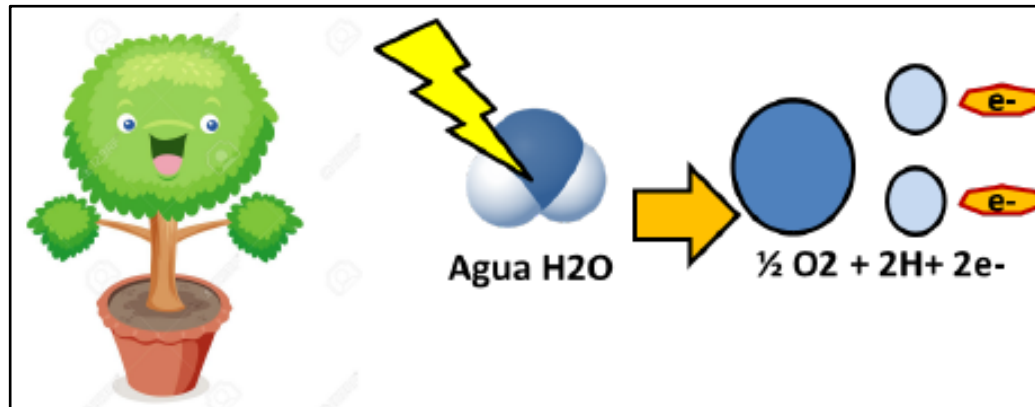
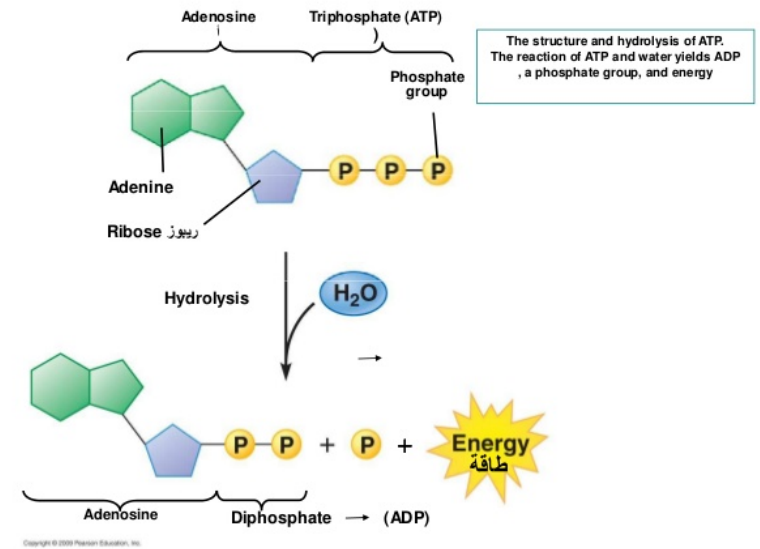
Also See: [https://en.wikipedia.org/wiki/Mineral_\(nutrient\)](https://en.wikipedia.org/wiki/Mineral_(nutrient))

Water

Polypeptide (or Protein) Folds in 3D



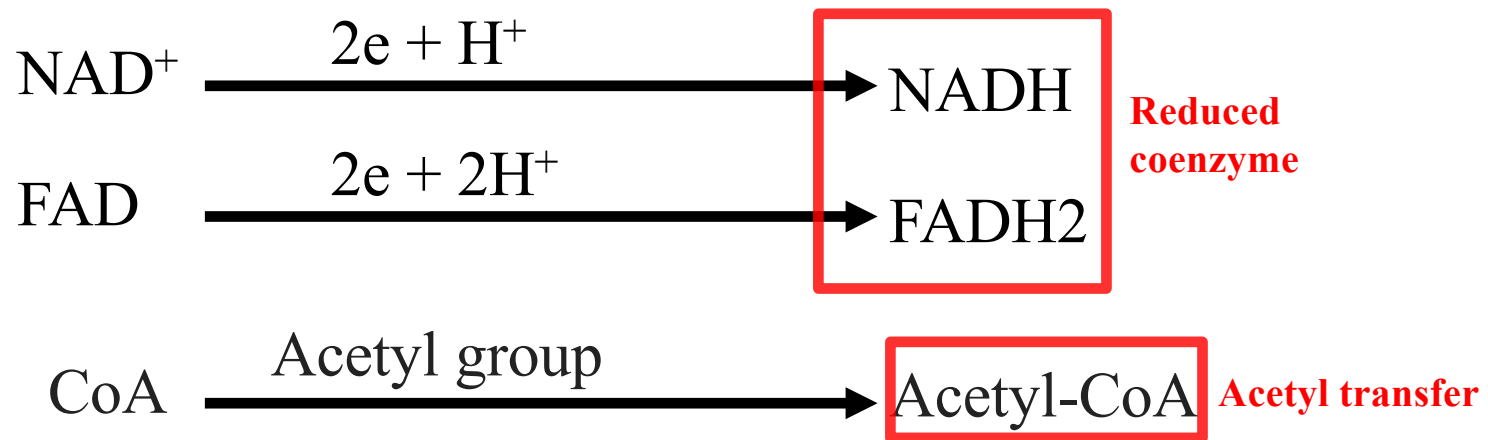
In water, hydrophobic residues (shown as black circles) tend to hide from water, thereby spontaneously folding protein to its native structure.



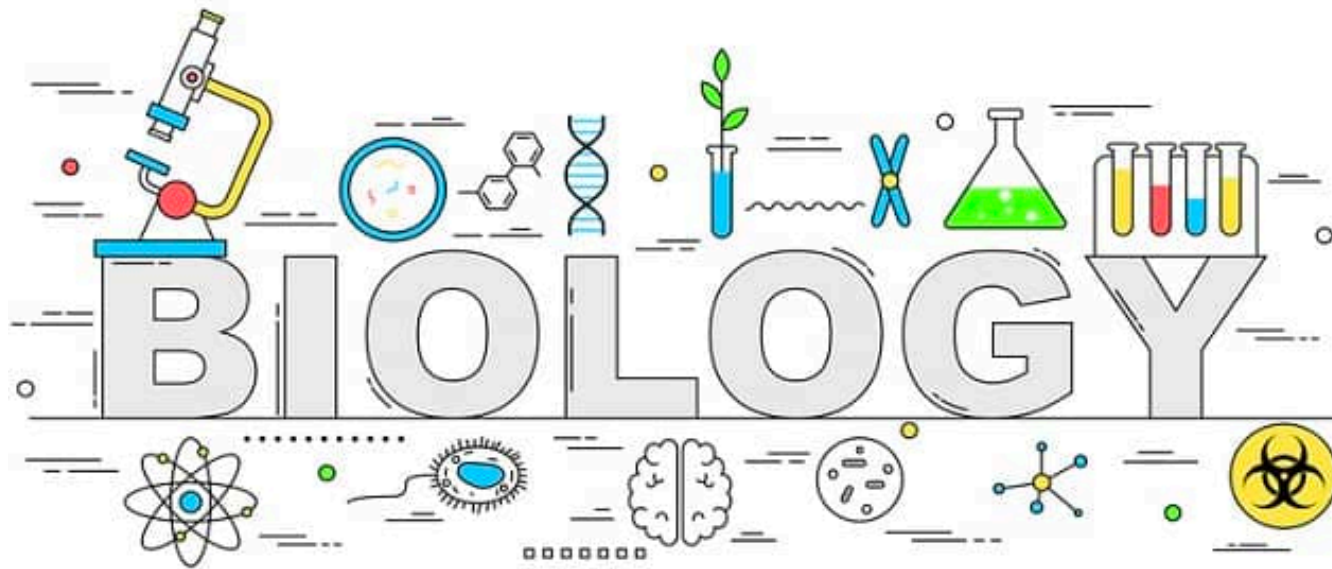
<https://www.youtube.com/watch?v=p9UC0chfXcg>

Take home

- **Vitamins** : Essential **organic nutrients**, required in **small amounts**.
- Fat soluble & Water soluble
- **Vitamins** produces **coenzymes** (**NAD⁺**, **FAD**, **CoA**)



- **Vitamin-D** is synthesized from Cholesterol and stored in fat. Technically not a vitamin.
- Mineral is a chemical element (*Macrominerals*, *Microminerals*)
- Water : Essential for life: photosynthesis, solvent, chemistry involving waters



Next: Glucose metabolism